

Budget Golf Launch Monitor — V1 Build Proposal

Rian · August 2026 · Hardware budget under \$150 · Must work indoors into a net and outdoors on a range

WHAT I AM BUILDING

Two sensors watching the same swing. A \$8 24 GHz Doppler radar sits 1.4 m behind the ball and measures **ball speed** and **clubhead speed**. A club-mounted **IMU**, which I already have, measures clubhead speed a second time by different physics, and gives swing tempo and impact timing. Carry comes from a trajectory model I fitted to TrackMan's published tour averages.

The two channels are the point: clubhead speed gets measured twice by independent means, and I can report which agrees better with a reference. I have not found that comparison published for parts at this price.

WHY RADAR IS THE PRIMARY SENSOR

- **The IMU cannot see the ball.** Ball speed, smash factor and carry must come from the radar. An IMU-only device guesses smash factor, and being 0.1 out is a 23-yard carry error.
- **Cameras cannot work outdoors at this budget.** PiTrac, the main open-source Raspberry Pi launch monitor, strobes infrared into a darkened scene; its authors state it is not an outdoor system, and it is a \$250–400 build.
- **Ball speed is the metric worth having.** My model reproduces TrackMan's published carry to 3.9% RMS across six reference shots, and shows carry tracking ball speed almost one to one: **1% of speed error gives 1.1% of carry error.**

WHAT IT WILL REPORT

METRIC	SOURCE	TARGET
Ball speed	Radar, measured	±2–4 mph
Clubhead speed	Radar, measured	±3–6 mph
Clubhead speed	IMU, measured	±9–10 mph
Tempo, impact timing	IMU, measured	—
Smash factor	Derived	±0.05–0.10
Carry / total / apex	Modelled	±8–15 yd
Launch angle	Phase 5, phone camera	±1–1.5°
Direction, spin	Not reported	—

One Doppler receiver gives one number. Launch direction needs two or more receivers, spin axis three or four.

HOW EACH CHANNEL WORKS

Radar. The mixer turns the Doppler shift into an audio tone, 161 Hz per m/s at 24 GHz, so a 167 mph ball gives 12 kHz. I amplify about 80 dB across 200 Hz to 15 kHz, record through a USB sound card and run a short-time Fourier transform. The clubhead is a broad ridge before impact and the ball a narrower one after, with the IMU marking the split.

IMU. Clubhead speed is angular velocity times effective radius, $v = \omega r$. The gyroscope gives ω , and the radius comes from club length, so the app must know which club is in play. Length varies about 30% across a set, which is why published IMU tests show the error changing sign between wedge and wood. A commercial glove sensor tested against TrackMan gave ±9–10 mph limits of agreement on single swings and about 12% random error. That is the number to beat.

THE CHECK I HAVE TO DO FIRST

Published motion capture puts peak club angular velocity at $2,096 \pm 239$ °/s. Most cheap gyroscopes stop at ±2,000 dps, so the mean sits above the rail and around two-thirds of full swings would clip, removing the exact peak that sets the answer and biasing speed low with no error flag.

Week 1: identify the part, log raw gyro at full rate, count samples at the limit. **If over 20% of swings clip:** move to the lead glove, where the published peak is $1,627 \pm 230$ °/s, or interpolate through the clipped peak.

PARTS

ITEM	COST
CDM324 24.125 GHz Doppler module	\$8.05
Two-stage OPA2365 amplifier, passives	~\$18
USB sound card, 48 kHz (the ADC)	~\$9
Filtered 5 V supply	~\$5
Perfboard, shielded cable, connectors	~\$12
Printed case, horn, tripod mount, club clip	~\$21
IMU and microcontroller	already have
Total	\$73–77

HOW I WILL PROVE IT WORKS

Two hours in a Golf Galaxy bay at \$49.99/hour, which runs TrackMan, arranged in advance so both sensors sit alongside. All three capture the same shot.

- Driver, 7-iron and pitching wedge, 30 shots each, over two sessions on separate days. Both channels are club-dependent, so a driver-only test proves nothing.
- Mean bias and 95% limits of agreement per metric per club, following Bland and Altman, plus the share of shots giving no reading.
- **Headline comparison:** radar clubhead speed against IMU clubhead speed, both against TrackMan, on identical swings.
- **Pass mark, set in advance:** radar ball speed within ±3 mph on the 7-iron and ±4 mph on the driver, 90% of shots detected.

MAIN RISKS

- **Radar signal strength.** A golf ball reflects roughly 30 dB less than a person, so the module's advertised 15 m range does not apply. Mounting close plus a printed horn is the answer. Most likely risk to sink the project, measured in week 4.
- **Gyroscope saturation.** A silent failure, so the clip count gets logged on every shot rather than checked once. The accelerometer saturates regardless, since a grip sensor sees around 82 g from the swing alone against ±16 g on a typical part, so speed comes from the gyroscope only.
- **Amplifier bandwidth.** Published circuits for these modules roll off near 1 kHz, having been built for pedestrians and cars. Copying one unmodified would cap me at 14 mph, so the passband needs redesigning and bench-sweeping.

PLAN

PHASE	WORK	WEEKS
P0	Python pipeline on synthetic data; IMU clip check	1–2
P1	Breadboard amplifier; IMU logging firmware	2–3
P2	Both sensors boxed, 100 shots into a net	2–3
P3	TrackMan sessions	2
P4	Cosine fit, radius fit, trajectory model, analysis	2–3
Total		11–15

WHAT I WOULD LIKE TO ASK YOU

- Does the university have a TrackMan or GCQuad I could use instead of paying for retail bay time?
- Can I get bench time with a signal generator and scope for P1?
- If the gyroscope clips on most swings, is glove mounting an acceptable change of scope, or should I spend \$6 on a ±4,000 dps part and keep it on the club?